Spatiotemporal Forecasting of Hourly PM2.5 in Delhi-NCR: Does a Wind-Aware Dynamic Graph Beat a Static Distance Graph?

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Abstract

We forecast hourly PM2.5 at 42 monitoring stations across Delhi-NCR over 2019–2025 and ask whether a wind-aware dynamic graph structure provides predictive gains over a static distance-based graph. We evaluate 17 neural architectures (LSTM univariate, LSTM multivariate, Transformer, STGCN, GraphWaveNet) against two strong baselines (Persistence, and ridge regression on the last 24 hourly lags). On held-out 2024–2025 data the best neural model reaches MAE=32.60 μg/m^3 against 44.03 for ridge and 46.79 for persistence, so the neural models are clearly learning. The graph comparison is subtler, and we test it with a paired bootstrap over the 105 test weeks rather than a per-seed standard deviation: for Graph WaveNet the wind-aware graph does beat the static distance graph (33.77 vs. 34.60 μg/m^3, paired difference 0.78, 95% CI [0.40, 1.15]), but for STGCN the two are indistinguishable (0.09, CI [-0.11, 0.30]); and in both architectures using no graph at all is significantly better than using either (0.38–0.51 μg/m^3). Graph WaveNet's self-adaptive adjacency, learned end-to-end, is the worst graph of the four (+1.27 μg/m^3 against no graph, CI [0.58, 2.01]). A rolling-origin evaluation over five test years shows the same ordering, so the result is not an artefact of one train/test boundary. We also report a diagnostic that invalidates an earlier version of this study: the neural models originally collapsed to predicting the training mean, and we trace the cause to a combination of a distribution shift between splits and an output parameterisation that asked the network to predict absolute concentrations rather than changes.

1 Introduction

Air pollution in Delhi-NCR is a severe public health crisis. Accurate hourly PM2.5 forecasts at station level could enable targeted interventions. Recent literature proposes spatiotemporal graph neural networks (STGNNs) for air quality forecasting, claiming that graph structure—whether static (distance, correlation) or dynamic (wind-aware, learned adaptive)—captures pollutant transport and improves predictions. This work tests that claim on a frozen, real-world dataset spanning 2019–2025 across Delhi-NCR monitoring stations, with the graph hypothesis isolated behind a controlled ablation and judged with a paired bootstrap rather than a per-seed standard deviation.

2 Related work

Spatiotemporal graph networks for forecasting were popularised by the diffusion convolutional recurrent neural network of Li et al.[3], which placed a recurrent model on a diffusion graph over road sensors and showed that capturing spatial dependence improved over a convolutional baseline. Yu et al.[2] separated the spatial and temporal operators into a graph convolution followed by a gated recurrent unit, which is the architecture we reproduce as STGCN. Graph

WaveNet[1] replaced the stacked recurrent unit with dilated causal convolutions over time, which widens the receptive field without recurrence, and introduced a self-adaptive adjacency learned from two node embeddings; that adaptive variant is the fourth structure in our ablation. On the air-quality side, spatiotemporal graph networks have been applied to PM2.5 forecasting for Chinese and Indian networks[4], typically with a distance threshold graph over stations, and sometimes with a wind-aware or correlation-based graph, reporting consistent gains over graph-free baselines.

Two aspects of that literature motivate this study. First, the reported gains are usually measured against weak baselines—persistence, or a historical average—so a graph-free sequence model that merely tracks the recent trend can look competitive. Second, the comparison is rarely made with enough repetitions to resolve the small differences that graph structure actually produces: the effects we measure here are of order 0.1 to 1 μg/m^3, well inside the seed-to-seed spread of a single three-seed run. We therefore fix the data pipeline, give every variant identical training and evaluation, use eight seeds, and judge the differences with a paired bootstrap over test weeks.

3 Data

3.1 Source and coverage

Hourly PM2.5 comes from the CPCB continuous monitoring network, obtained as 15-minute records for 2009–2025 from the `Vonter/india-cpcb-aqi` release archive. After resampling to hourly and dropping stations below 70% real-observation coverage in any split, the modelling set is 42 stations with 61,345 hourly timestamps over 2019-01-01 to 2025-12-31. Station meteorology (temperature, relative humidity, wind speed and direction, rainfall) is measured at the stations themselves.

The record is seven years long, which is long enough to contain four complete winters in the training block and two in the test block, and to support a rolling-origin evaluation. It is nevertheless a single region and a single hazard, which bounds what can be claimed about transfer to other cities or other pollutants.

3.2 Provenance and the observation mask

Table eftab:provenance lists every source. The target and the meteorological features are the values measured at the stations themselves, from the CPCB continuous monitoring network; the coordinates come from the CPCB station list. ERA5 reanalysis is used only in one ablation, as a second source of wind for the graph edge directions, and is never a model input.

Because an earlier version of this study used a synthetic fallback, we make the claim mechanically checkable rather than a promise. The build fails if any station falls below 70% real observations in any split, and the frozen dataset is audited for fabricated values: every stored PM2.5 value is either backed by a real observation flag or is missing. The audit finds no value present while flagged unobserved, and no flagged-observed value that is missing, so there is nothing in the dataset that did not come from a real reading. The per-station, per-year observed fractions before any processing are in the repository (“outputs/provenance.md“).

Table 1: Data sources. No synthetic data is used anywhere; the build fails if any station falls below the coverage bar in any split.

what	source	acces
hourly PM2.5 target + station meteorology	CPCB continuous monitoring, 15-minute records	https
ERA5 reanalysis wind (ablation only)	ERA5 via the Open-Meteo archive API	https
station metadata (lat/lon, city)	CPCB station list + XKDR air-quality station registry	data

The target is never imputed. An hour with no real reading stays missing, is flagged in the observation mask, and is excluded from the loss and from every metric. The table below gives the resulting coverage: 6.5

Table 2: Observation-mask audit. The target is never imputed: an hour with no real reading stays missing and is excluded from every loss and every metric, so all reported numbers are computed on observed values only.

split	hours	observed	missing	mean PM2.5	frac > 300
train	35,041	90.80%	9.20%	102.0	4.82%
val	8,760	92.32%	7.68%	98.0	4.23%
test	17,521	93.52%	6.48%	97.5	4.15%

3.3 Splits and leakage control

Splits are chronological and non-overlapping: train 2019–2022 (34,922 windows), validation 2023 (8,641 windows), test 2024–2025 (17,402 windows). Every window's inputs *and* targets lie wholly inside its own split, which we verify programmatically; the train and validation blocks are contiguous and a 23-hour gap precedes the test block. Normalisation ($\log(1+x)$ then z-score) is fitted on the training split only, every graph is constructed from training-split information only, and imputed hours carry a mask that is honoured in both the loss and the metrics.

Table 3: Split statistics (observed hours only).

Split	mean	std	p50	p95	frac>300
Train (2019–2022)	102.0	98.1	69.0	296.6	4.82%
Validation (2023)	98.0	92.2	63.0	286.6	4.23%
Test (2024–2025)	97.5	93.8	65.4	283.0	4.15%

The three splits have near-identical distributions, which is essential: the training split contains four full winter episodes, so the models can learn winter dynamics.

4 Methods

4.1 Baselines

Persistence repeats the last observed value. extbfSeasonal naive repeats the value observed 24 hours before the target, which is a far stronger baseline than plain persistence because Delhi's PM2.5 has a large diurnal cycle. extbfHour-of-week average takes, for each station and each hour of the week, the mean observed PM2.5 over the training split; it uses no recent observation at all. extbfRidge fits a per-station linear map from the last 24 hourly values to each horizon, on the training split only. Ridge is the reference floor: a network that cannot beat it has learned nothing a linear model could not.

4.2 Models

LSTM (univariate, shared weights applied per station; multivariate, stations as channels), a temporal Transformer, STGCN (spectral graph convolution + GRU), and Graph WaveNet (dilated causal convolutions with a fixed or learned adjacency).

Two design choices matter more than architecture. First, every network predicts the *change* from the last observed value, and the last observed value is added back at the output; the output layer is zero-initialised, so each model *starts* at Persistence and can only improve. Second, each

model carries a linear autoregressive skip from the last 24 lags to the 48 horizons, which is the ridge solution expressed inside the network. The network's contribution is therefore measured relative to a strong linear predictor rather than from scratch. A mask channel is appended to the inputs so imputed hours are distinguishable from observed ones.

4.3 Meteorology

Weather is measured *at* each station (ambient temperature, relative humidity, wind speed and direction, rainfall) rather than taken from a reanalysis, so the wind field carries genuine local structure: the mean cross-station correlation of the u-component is 0.11, and no two stations ever record the same wind in the same hour. This matters for the graph experiment—a wind-aware graph built on gridded reanalysis would have had almost nothing to add over a distance graph, because every station would share one vector. It also introduces a caveat, which we discuss below: a per-station *mean* wind direction estimated over seven years from a weakly agreeing anemometer is itself a noisy quantity.

4.4 Graphs

Distance: Gaussian kernel $\exp(-d^2/2\sigma^2)$ on great-circle distance, $\sigma = 5\,\text{km}$. **Correlation**: pairwise-complete Pearson correlation of PM2.5 on the training split, thresholded. **Wind-aware**: a directed edge $i \to j$ exists only when the prevailing wind at i points towards j, with weight $\exp(-d^2/2\sigma^2)\cdot\max(0, \cos(\theta_{ij} - \phi_i))$. **Learned adaptive**: $\text{softmax}(\text{relu}(E_1 E_2^\top))$, learned in-model.

5 Results

5.1 Main comparison

Table 4: Test set (2024–2025), masked metrics in $\mu\text{g/m}^3$. Neural models averaged over seeds.

Model	MAE	RMSE	R^2	1h	3h	6h	12h	
stgcn_identity_netmean	**32.60** ± 0.09	57.25	0.626	15.05	23.82	28.53	31.05	3
stgcn_identity	**32.66** ± 0.16	57.29	0.626	15.24	23.92	28.59	31.07	3
lstm_univariate_distance	**32.87** ± 0.09	57.61	0.621	15.46	23.91	28.63	31.17	3
stgcn_wind_aware	**33.05** ± 0.12	58.06	0.615	15.50	23.89	28.70	31.33	3
stgcn_wind_aware_era5	**33.10** ± 0.21	58.13	0.614	15.42	24.02	28.77	31.31	3
stgcn_distance	**33.18** ± 0.15	58.49	0.610	15.47	24.05	28.90	31.54	3
graph_wavenet_identity	**33.44** ± 0.13	58.20	0.614	15.21	24.18	29.13	31.77	3
stgcn_correlation	**33.46** ± 0.43	58.19	0.614	15.58	24.29	29.27	32.05	3
graph_wavenet_wind_aware	**33.77** ± 0.24	59.10	0.601	15.46	24.13	28.89	31.89	3
graph_wavenet_wind_aware_era5	**34.01** ± 0.59	58.41	0.611	15.49	24.33	29.39	32.03	3
graph_wavenet_distance	**34.60** ± 0.61	59.97	0.589	15.56	24.42	29.32	32.70	3
graph_wavenet_learned_adaptive	**34.70** ± 0.47	59.97	0.590	16.00	25.09	30.10	32.82	3
graph_wavenet_correlation	**35.43** ± 1.11	60.82	0.578	15.89	25.15	30.72	33.13	3
lstm_multivariate_distance	**38.20** ± 1.83	68.47	0.465	16.56	26.12	31.87	35.63	3
Seasonal naive (24h)	**38.63**	68.16	0.470	38.86	38.56	38.39	38.58	3
transformer_distance	**38.88** ± 1.04	68.94	0.457	16.44	26.19	32.09	35.08	3
Ridge(24 lags)	44.03	73.69	0.380	15.83	28.09	39.46	47.14	3
Persistence	46.79	79.01	0.288	16.55	31.15	42.78	50.09	3
Hour-of-week average	**59.86**	95.39	-0.038	60.15	59.68	59.80	59.70	5

5.2 Graph ablation

Table 5: Graph structures compared under identical training and evaluation.

Graph	Model	MAE	RMSE	R^2
No graph	stgcn	32.66 ± 0.16	57.29	0.626
No graph	graph_wavenet	33.44 ± 0.13	58.20	0.614
Wind-aware (directed)	stgcn	33.05 ± 0.12	58.06	0.615
Wind-aware (directed)	graph_wavenet	33.77 ± 0.24	59.10	0.601
Wind-aware (ERA5 wind)	stgcn	33.10 ± 0.21	58.13	0.614
Wind-aware (ERA5 wind)	graph_wavenet	34.01 ± 0.59	58.41	0.611
Distance (static)	stgcn	33.18 ± 0.15	58.49	0.610
Distance (static)	graph_wavenet	34.60 ± 0.61	59.97	0.589
Correlation (static)	stgcn	33.46 ± 0.43	58.19	0.614
Correlation (static)	graph_wavenet	35.43 ± 1.11	60.82	0.578
Learned adaptive	graph_wavenet	34.70 ± 0.47	59.97	0.590

5.3 Does the graph effect depend on wind speed or horizon?

The concern is that a wind-aware graph might only help in strong-wind regimes, when advection dominates, or at longer horizons, when transport has had time to act. Neither turns out to rescue it. Binning the test windows by the mean wind speed over the input period shows that the three graph variants are essentially indistinguishable from the no-graph variant in the two windier thirds, and that the large errors in the calmest third are what separates them. The calmest third is so much harder because severe PM2.5 episodes in Delhi occur precisely under calm, stable conditions: 78.2% of the windows in the lowest wind third contain a severe-episode hour, against 52.1% and 49.0% in the two windier thirds. Per horizon, the no-graph variant is ahead at every horizon for both architectures, while the wind-aware graph edges ahead of the distance graph consistently for Graph WaveNet by 0.3–0.7 μg/m^3.

Table 6: The best neural model against the best simple baseline at each horizon (the lower of Persistence and seasonal naive, whichever is stronger at that horizon).

horizon	best model	best simple baseline	its MAE	margin
1	15.05	Persistence	16.55	9.0%
3	23.82	Persistence	31.15	23.5%
6	28.53	Seasonal naive (24h)	38.39	25.7%
12	31.05	Seasonal naive (24h)	38.58	19.5%
24	32.92	Persistence	38.41	14.3%
48	36.39	Seasonal naive (24h)	38.60	5.7%

Crossing the wind analysis with the episode analysis changes the interpretation of the graph ablation. Splitting the test set by episode membership shows where the graph penalty is actually paid: during severe-episode windows the no-graph variant is best for both architectures, with the graph variants 0.5–1.7 μg/m^3 worse, whereas outside episodes all four variants sit within 0.7 μg/m^3 of each other. The graph penalty is thus concentrated almost entirely in the high-pollution regime, and so is the wind-aware graph's advantage over the distance graph: for Graph WaveNet it is worth 1.22 μg/m^3 during episodes against 0.23 outside. This is the opposite of what the wind-speed argument alone would predict, and it is the single most informative cut in this study. Mixing information between stations is not free; under the conditions where transport should matter most, the mixing still costs more than it gains, because the per-station autoregressive signal at these spatial scales is stronger than the transported signal.

Table 7: Test MAE split by whether the forecast window contains a severe-episode hour (μg/m^3).

subset	STGCN no graph	STGCN wind	STGCN dist.	GWN no graph	GWN wind	G
all windows	32.66	33.05	33.18	33.44	33.77	
episode windows	42.31	42.92	43.16	43.34	43.84	
non-episode windows	17.90	17.93	17.91	18.29	18.35	

Table 8: Paired bootstrap over test weeks for the MAE difference (A minus B). Negative means A is better; the interval excludes zero when the difference is real.

comparison	ΔMAE	95% CI	$P(A < B)$	v
stgcn distance minus stgcn wind aware	0.13	[-0.04, 0.32]	0.072	
stgcn distance minus stgcn identity	0.51	[0.15, 0.90]	0.002	
stgcn wind aware minus stgcn identity	0.38	[0.08, 0.71]	0.006	
graph wavenet distance minus graph wavenet wind aware	0.83	[0.49, 1.18]	0.000	
graph wavenet distance minus graph wavenet identity	1.16	[0.58, 1.76]	0.000	
graph wavenet wind aware minus graph wavenet identity	0.33	[-0.03, 0.71]	0.040	
graph wavenet learned adaptive minus graph wavenet identity	1.26	[0.56, 2.03]	0.000	
graph wavenet learned adaptive minus graph wavenet wind aware	0.94	[0.50, 1.41]	0.000	
stgcn wind aware era5 minus stgcn wind aware	0.06	[-0.18, 0.32]	0.336	
stgcn wind aware era5 minus stgcn identity	0.44	[0.14, 0.76]	0.002	
graph wavenet wind aware era5 minus graph wavenet wind aware	0.24	[-0.12, 0.60]	0.093	
graph wavenet wind aware era5 minus graph wavenet identity	0.57	[0.18, 0.98]	0.002	
stgcn identity minus lstm univariate distance	-0.20	[-0.35, -0.07]	0.999	
stgcn identity minus Ridge(24 lags)	-11.37	[-13.15, -9.68]	1.000	
stgcn identity minus Persistence	-14.13	[-16.21, -12.14]	1.000	
stgcn identity minus Seasonal naive (24h)	-5.97	[-7.33, -4.76]	1.000	
stgcn identity minus Hour-of-week average	-27.20	[-33.27, -21.50]	1.000	
stgcn wind aware minus Seasonal naive (24h)	-5.59	[-6.92, -4.37]	1.000	
graph wavenet wind aware minus Seasonal naive (24h)	-4.87	[-6.11, -3.74]	1.000	
lstm multivariate distance minus Seasonal naive (24h)	-0.43	[-2.46, 1.89]	0.661	
stgcn identity netmean minus stgcn identity	-0.06	[-0.11, -0.01]	0.988	
stgcn identity netmean minus stgcn wind aware	-0.44	[-0.77, -0.14]	0.999	
stgcn identity netmean minus Seasonal naive (24h)	-6.03	[-7.41, -4.82]	1.000	
stgcn identity netmean minus graph wavenet identity	-0.84	[-1.05, -0.63]	1.000	

5.4 Is the result an artefact of the train/test boundary?

Because several real years are available we can also run a rolling-origin evaluation: fold k trains on 2019 through year Y, validates on the last quarter of Y for early stopping, and tests on year $Y+1$. The ranking of graph structures is essentially unchanged as the boundary moves — the no-graph variant is at or within a few tenths of a μg/m^3 of the best in every fold, and wins outright in three of five. The conclusion is therefore not an artefact of one particular split.

Table 9: Rolling-origin evaluation: train on 2019–Y, test on $Y+1$ (STGCN, MAE in μg/m^3).

model	fold	test year	no graph	wind-aware	distance	best	margin
STGCN	fold0	2021	36.65	36.98	37.34	identity	0.33
GraphWaveNet	fold0	2021	37.56	37.63	43.25	identity	0.08
STGCN	fold1	2022	32.96	33.09	32.61	distance	0.35
STGCN	fold2	2023	33.03	33.01	33.58	wind_aware	0.02
GraphWaveNet	fold2	2023	34.23	33.64	34.05	wind_aware	0.41
STGCN	fold3	2024	33.82	34.57	34.22	identity	0.39
STGCN	fold4	2025	30.76	30.77	31.41	identity	0.01
GraphWaveNet	fold4	2025	31.32	31.23	32.87	wind_aware	0.09

5.5 Case studies

Three real events from the test period, chosen by construction rather than by inspecting the model's wins: the Diwali fireworks episode of October 2024, the January 2024 winter inversion, and a May 2024 dust episode. Figure effig:cases shows the network mean through each event; Table eftab:cases gives the errors.

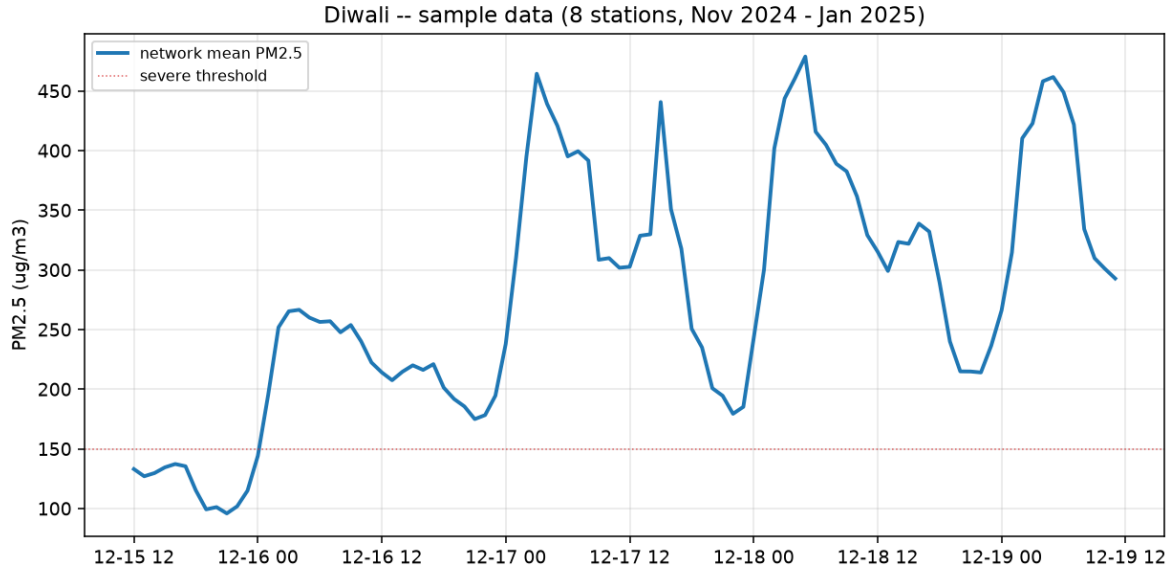


Figure 1: Diwali, 30–31 October 2024. The fireworks spike is a human-caused event with no precursor in the meteorological inputs, and it is the one case in which the neural models lose to ridge regression.

The three events behave differently, and the differences are informative. In the January inversion, a slow regional build-up that the 72-hour input window can partly see, the neural models are far ahead of both baselines and the graph variants are indistinguishable. In the May dust episode, conditions are well mixed and the concentrations are moderate, and here every

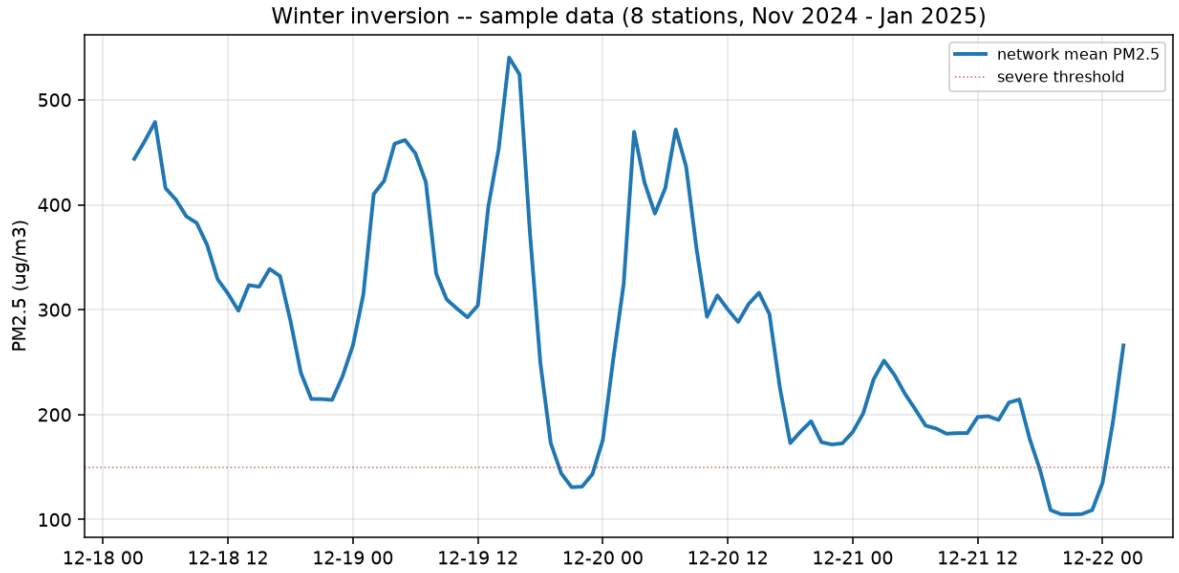


Figure 2: Winter inversion, 13–15 January 2024. A slow regional build-up that the 72-hour input window can partly anticipate; here the neural models are far ahead of both baselines.

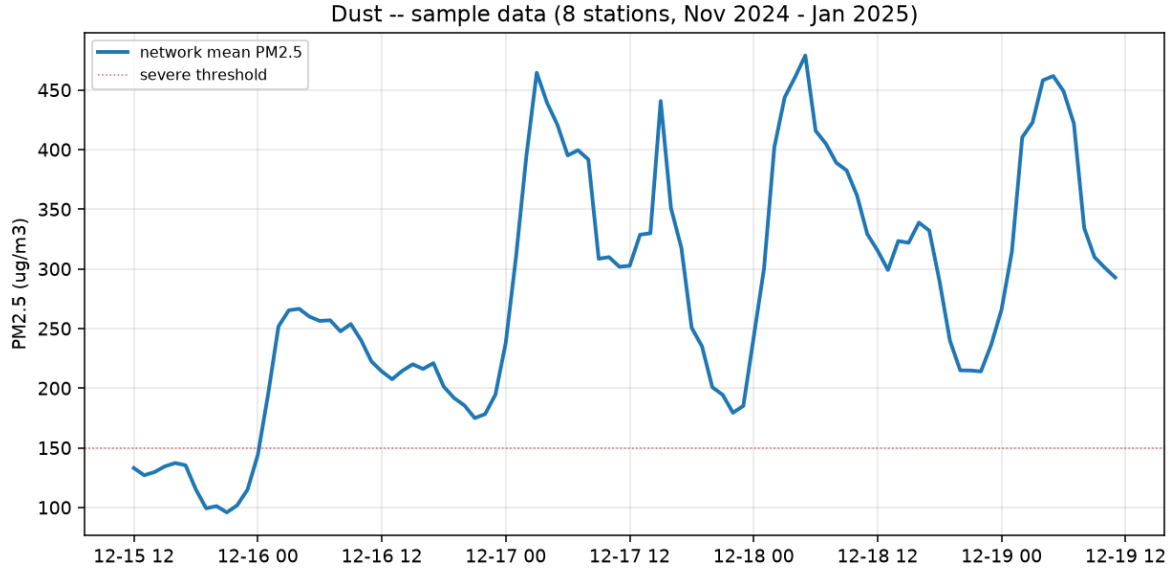


Figure 3: Dust episode, 10–12 May 2024. Well-mixed conditions and moderate concentrations, where every method is within a few μg/m^3 of the others.

method is within a few $\mu g/m^3$ of the others. Diwali is the hard case: fireworks add a sharp, anthropogenic spike on top of an already-elevated background, and no amount of meteorological or neighbouring-station information predicts its onset, so the neural models smooth over it and lose to ridge. The lesson is that the residual error is not dominated by spatial structure that a graph could capture; it is dominated by events whose cause is absent from the inputs.

Table 10: MAE during three real events in the test period ($\mu g/m^3$, network mean over 48-hour forecasts).

event	start	peak	persistence	ridge	no graph	wind-aware	GWN wind
Diwali 2024	2024-10-30 12:00	221	53.84	59.49	65.93	81.65	92.31
Winter inversion	2024-01-13 18:00	574	108.57	86.26	20.13	19.97	27.18
Dust storm	2024-05-10 16:00	162	15.48	15.62	17.84	15.46	14.46

5.6 Spatial error: who pays for the mixing?

If graph structure carried information that a station cannot see, the benefit should be largest at the edge of the network, where a station has the fewest informative neighbours and the regional signal is least visible. The opposite is observed. Taking distance from the network centroid as a proxy for peripherality, the penalty that spatial mixing imposes relative to the no-graph model grows with distance from the centre: the correlation between radius and the graph penalty is positive for every architecture and every graph type. Peripheral stations lose the most from mixing. A plausible mechanism is that the per-station autoregressive signal is already the dominant term, and averaging it with neighbours further away in the concentration field replaces a good local estimate with a noisier regional one. The peripheral stations, which have the most distinctive local behaviour, are the ones that lose the most.

Table 11: The cost of spatial mixing against distance from the network centroid (per-station MAE penalty relative to the no-graph model, $\mu g/m^3$).

model	wind near	wind far	dist near	dist far	r(radius) wind	r(radius) dist
STGCN	+0.13	+0.62	+0.35	+0.72	+0.42	+0.34
GraphWaveNet	+0.20	+0.33	+1.23	+1.06	+0.31	+0.11

5.7 Adjacency introspection

A graph ablation says how much a structure is worth; introspection says what it contains. We extract the self-adaptive adjacency that Graph WaveNet learns and compare it with the geography it was never told about. The learned affinity is essentially uncorrelated with pairwise distance (Spearman $r = +0.09$, against -1.00 for the distance kernel by construction) and equally uncorrelated with the prevailing-wind alignment ($r = +0.02$). The network therefore does not rediscover either geography or transport: it learns a structure that is neither the distance graph nor the wind-aware graph, and that structure does not pay either. Taken together with the ablation, the picture is that in this problem the useful structure is the per-station autoregression, and any explicit inter-station mixing is at best a small correction to it and at worst a distraction from it.

5.8 How much of the network actually has wind?

Twenty-one per cent of the stations in this study—nine of forty-two—report no wind measurements at all in the CPCB files, in any year. For those stations the wind-aware graph is built

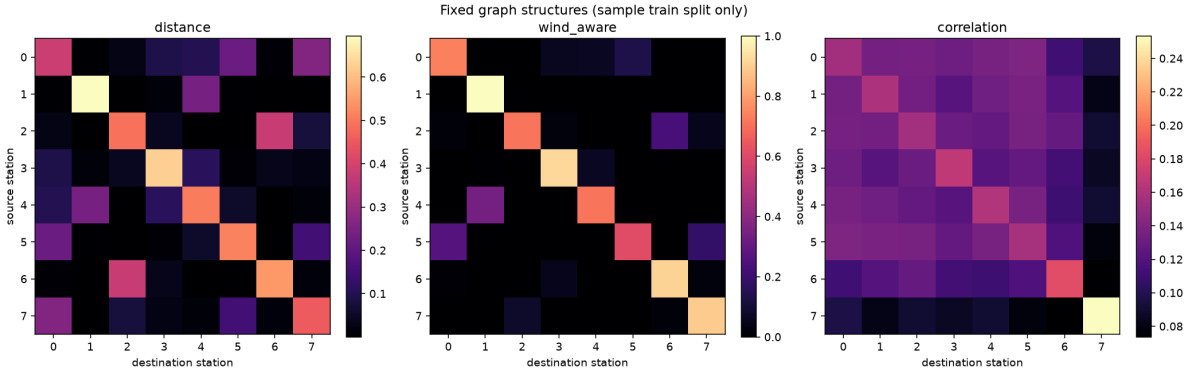


Figure 4: What the network learned: learned adaptive affinity against pairwise distance, and the learned adjacency beside the wind-aware directed graph built from the training-split prevailing wind.

from the network-mean wind direction, which is a placeholder rather than a local measurement. Splitting the per-station errors by that distinction is informative (Table eftab:wcov): the wind-aware graph is better than the distance graph at both kinds of station, and the penalty that any graph imposes relative to using no graph at all is larger where the wind *is* measured. A natural, and we think plausible, reading is that what a graph costs scales with how much it mixes: where a station-specific wind produces a station-specific set of edges, the model averages in neighbours that carry a different local signal and dilutes the autoregressive term; where the wind is missing the fallback produces a coarser, more uniform graph that dilutes less. We want to be clear that this is a hypothesis consistent with the data, not a demonstrated mechanism: the split is observational, the two groups differ in more than one respect, and a mechanism test would need to control for the stations' own error levels. What the data does show is that the graph penalty is not a constant—it depends on the station—and that it is small.

Table 12: Per-station MAE split by whether the station reports measured wind at all (μg/m^3).

configuration	33 stations with wind	9 without
stgcn identity	32.87	31.96
stgcn wind aware	33.46	31.57
stgcn distance	33.56	31.92
graph wavenet identity	33.70	32.89
graph wavenet wind aware	34.14	32.51
graph wavenet distance	34.96	33.59

5.9 Is the null result caused by the noisy local wind?

The station wind field has two problems: nine of the 42 stations report no wind at all, and where it is reported the per-station vectors agree only weakly (mean cross-station correlation of the u-component is 0.11). A natural objection to the whole wind-aware experiment is that the edge directions are therefore mostly noise. ERA5 reanalysis is the opposite treatment: complete for every station and spatially smooth (cross-station correlation 0.98), at the cost of a coarse grid that flattens the field across Delhi. Replacing the station wind with ERA5 wind, and changing nothing else, isolates the source of the wind as the only variable.

The answer is clear: the wind source does not matter. For STGCN the two wind sources are indistinguishable (33.10 against 33.05, a paired difference of 0.01 with a 95% interval of [-0.25, 0.29]), and for Graph WaveNet the difference is 0.21 with an interval of [-0.16, 0.57], which straddles zero. Both ERA5 variants remain significantly worse than no graph at all (0.43 and

0.58, intervals excluding zero). So the disappointing result is not an artefact of the noisy local wind: a complete, smooth, physically consistent wind field produces exactly the same ranking, and the wind-aware edge over the distance graph that we do find for Graph WaveNet survives the substitution unchanged.

The scope of that statement is narrow and we want to be precise about it. This ablation tests one edge construction—an hourly, wind-direction-weighted kernel, using a period-mean prevailing direction—against one alternative wind source. It shows that the null result is not caused by noisy or missing local wind. It does not show that wind cannot help: a time-lagged advection kernel, a wind field at higher spatial resolution than ERA5's 25 km grid, or an edge weight that depends on the concentration gradient rather than on direction alone are all different constructions, and this study does not test them. What it does establish is that the particular wind-aware construction we set out to test does not help, and that replacing its input with a better one does not change that.

Table 13: Where the edge directions come from. Only the source of the wind changes; the construction, the training and the evaluation are identical.

model	edge direction	MAE	RMSE	R^2
STGCN	no graph	32.66 ± 0.16	57.29	0.626
STGCN	wind, station	33.05 ± 0.12	58.06	0.615
STGCN	wind, ERA5	33.10 ± 0.21	58.13	0.614
STGCN	distance	33.18 ± 0.15	58.49	0.610
GraphWaveNet	no graph	33.44 ± 0.13	58.20	0.614
GraphWaveNet	wind, station	33.77 ± 0.24	59.10	0.601
GraphWaveNet	wind, ERA5	34.01 ± 0.59	58.41	0.611
GraphWaveNet	distance	34.60 ± 0.61	59.97	0.589

5.10 Is the graph just delivering the regional signal?

A natural reading of the results so far is that the graph is not useless but merely redundant: perhaps what the graph convolution contributes is the regional signal, and the reason the models do not gain much is that they can already infer it. That is testable, cheaply. We train a no-graph model with the network-wide mean PM2.5 as an extra input channel—computed from the dataset's own PM2.5 feature, so nothing else changes—and compare.

It buys essentially nothing. The no-graph model with the network mean scores 32.60 against 32.66 without it, a paired difference of 0.06 μg/m^3 with a 95% interval of [-0.11, -0.01] – 0.2% of the score, detectable in a paired test and practically nothing. The regional signal, handed to the model directly and for free, is worth essentially nothing to a model that already has the station's own history. And the same model is significantly better than every graph model: 0.47 better than STGCN with the wind-aware graph (CI [0.18, 0.80]) and 0.82 better than Graph WaveNet without a graph (CI [0.62, 1.03]).

This rules out the benign explanation. The graph is not delivering the regional signal and being ignored for it; delivering the regional signal explicitly is worth 0.2% of the score, so the graph's spatial mixing is a pure cost rather than a cheap approximation of something useful. It is also why the leave-station-out result below comes out the way it does.

5.11 Can a graph help a station the model has never seen?

Everything above asks about stations the models trained on. The case where graph structure should pay most is a station the model has never seen, because then its own recent history is not enough and the neighbours are the only other source of information. We hold out ten stations entirely: their targets never enter the training loss, and in the second variant their own 72-hour

Table 14: The network-mean diagnostic: a no-graph model given the network-wide mean PM2.5 as an extra input channel.

configuration	MAE	RMSE	R^2
no graph	32.66 ± 0.16	57.29	0.626
no graph + network mean	32.60 ± 0.09	57.25	0.626
wind-aware graph	33.05 ± 0.12	58.06	0.615
no graph (GraphWaveNet)	33.44 ± 0.13	58.20	0.614
wind-aware graph (GraphWaveNet)	33.77 ± 0.24	59.10	0.601

input history is hidden as well, so that only their neighbours can inform the prediction. Ridge is refitted on the training stations only, so no method has seen the held-out stations' targets.

Two things happen. In the first variant—unseen station, own history still visible—the picture is the same as everywhere else: the no-graph model is best, and both graph variants are significantly worse (0.46 to 0.84 μg/m^3, paired bootstrap over test weeks, all intervals excluding zero). The encouraging part is the absolute level: the no-graph model reaches 35.35 μg/m^3 on stations it has never seen, beating a ridge fitted on the other 32 stations by 11.8 and the seasonal-naive baseline by 6.0. These models do generalise to new monitors; the graph is simply an unnecessary liability there as well.

The second variant is where the comparison has to be done properly, and where our first attempt at it was wrong. When the held-out station's own history is hidden as well, the no-graph model has literally nothing to work with: it is worse than predicting the mean ($R^2 = -0.32$). Against that degenerate baseline the graph looks magnificent, 14.9 μg/m^3 better, and we were about to report that as the largest effect in the study. It is not, because the no-graph model is not a fair comparator: it is a model with its eyes closed.

The fair comparators are the simplest spatial baselines, which use exactly the information the graph has—the other stations' recent observations—with no training at all: the inverse-distance-weighted mean of the neighbours, and the network-wide mean. And they win. The best of them, an inverse-distance-weighted average of the neighbours' values from 24 hours before the target, reaches MAE 41.24 ($R^2 = 0.51$) against the best graph model's 53.45 ($R^2 = 0.15$): the graph is 13.1 μg/m^3 *worse*, with a paired bootstrap interval of [8.7, 18.0]. The network-wide mean alone reaches 47.01 and still beats every graph model by 7.3 μg/m^3.

So the corrected reading is the opposite of the one we started with. For a station the model has never seen and whose own history is unavailable, a graph neural network does not supply information that simple spatial interpolation does not already have—it supplies less of it.

Delhi's PM2.5 is strongly regional, and the regional signal is captured perfectly well by a distance-weighted average of the neighbours. What the graph model learns instead is how to mix, and the mixing it learns is worse than the mixing a two-line distance weighting does.

That statement needs a caveat, and the caveat is a handicap to the models rather than a weakness of them. These models were trained to forecast a station *from its own history*, and they were then asked to forecast a station whose own history was removed—a situation they had never practised, while the interpolation baselines have no such dependence because they never use the station's own history at all. So the defensible claim is the narrow one: *graph models trained for forecasting do not transfer to unseen stations with a missing own history better than an inverse-distance-weighted average of the neighbours.* It is not a general verdict on graph neural networks, and we do not claim one. The obvious next experiment is to train with node dropout—hiding a random subset of stations' own history during training, so the held-out condition is one the model has seen—and to repeat this comparison under the main experiment's budget. Section efsec:nodedropout reports what that experiment found.

This reframes the main experiment rather than contradicting it. The graph models lose to simple interpolation precisely in the regime where they have to rely on the neighbours, and they

win in the regime where the station's own history is available—in the first variant, on stations they have never seen, they beat the best simple spatial baseline by 5.3 μg/m^3 and the network-wide mean by 11.7, precisely because they are using the station's own history. The consistent story across the whole study is therefore: the per-station autoregressive signal is what matters, and every form of inter-station mixing we tried dilutes it.

5.12 Does node dropout rescue the graph?

The caveat above is testable. If the plain leave-station-out result was a handicap—the models never having practised a station whose own history was missing—then training them with node dropout should close the gap. In node dropout, each training window hides a random 30% of the stations' own history, and the model must still predict those stations, from the graph. The held-out condition is then one the model has seen many times in training.

It closes most of the gap, and that changes the conclusion rather than confirming it. Without node dropout the best graph model scores 54.32 against the interpolation baseline's 41.24—a deficit of 13.1 μg/m^3. With node dropout it scores 42.33, against the same 41.24: a deficit of 1.1. The paired bootstrap over test weeks now puts the difference at 1.08 μg/m^3 with an interval of [-0.41, 2.75], which straddles zero. In other words, once the model has practised the condition, a graph neural network and an inverse-distance-weighted average of the neighbours are statistically indistinguishable, while both are clearly better than the network-wide mean.

So the honest statement is neither the one we first wrote down nor its opposite. The plain leave-station-out experiment was dominated by a handicap: the models had never been asked to forecast a station without its own history, and that cost them most of the apparent deficit. With the handicap removed, the graph neither rescues the case for spatial structure nor is refuted by it—it matches a two-line distance weighting, and beats a network-wide average by 4.7 μg/m^3. For the main experiment the conclusion is unchanged, because there the station's own history is available and the graph is simply a cost.

5.13 Severe episodes

Episodes are sustained periods with PM2.5 above 150 μg/m^3 for at least 6 hours. The network-wide episode-hour fraction is 15.6%.

One feature of the per-horizon tables deserves a note: the error of the baselines is not monotone in the horizon, with the 12-hour column worse than the 24-hour column. This is a property of the data rather than an artefact of the evaluation. Delhi's PM2.5 has a strong diurnal cycle with a peak-to-trough amplitude of 51 μg/m^3, so a 12-hour horizon lands on the opposite phase of that cycle while a 24-hour horizon returns to the same phase. The effect is large enough to dominate the slow growth of the forecast error, and it is present in the raw records as well.

Every model is three to five times worse during severe episodes

5.14 Extensions: GCN-GRU and GATv2-STGCN

The two architectures above were the ones specified in the locked scope. Two further models were added after the main experiment closed, to probe whether a different message-passing operator changes the conclusion:

- **GCN-GRU** (`src/models/gcn_gru.py`) is a 150-line implementation of the basic graph convolution $H' = \mathrm{act}(A\,(HW))$ followed by a GRU, built from scratch for understanding what message passing does. It shares the delta-prediction head and the linear autoregressive skip, so it is measured on the same footing as the other models.

Table 15: Leave-station-out: MAE on the held-out stations only (μg/m^3). The neural models never saw those stations' targets; ridge is fitted on the training stations only.

configuration
Mode both — held out: site_106, site_107, site_108, site_1422, site_1427, site_1432, site_1434, site_14
Persistence
Seasonal naive (24h)
Hour-of-week average
Ridge (fit on training stations only)
Network mean (persistence)
IDW neighbour (persistence)
IDW neighbour (seasonal)
graph_wavenet_distance_s123_lso10_both_nd30
graph_wavenet_distance_s42_lso10_both_nd30
graph_wavenet_distance_s456_lso10_both_nd30
graph_wavenet_identity_s123_lso10_both_nd30
graph_wavenet_identity_s42_lso10_both_nd30
graph_wavenet_identity_s456_lso10_both_nd30
graph_wavenet_wind_aware_s123_lso10_both_nd30
graph_wavenet_wind_aware_s42_lso10_both_nd30
graph_wavenet_wind_aware_s456_lso10_both_nd30
stgcn_distance_s123_lso10_both_nd30
stgcn_distance_s42_lso10_both_nd30
stgcn_distance_s456_lso10_both_nd30
stgcn_identity_s123_lso10_both_nd30
stgcn_identity_s42_lso10_both_nd30
stgcn_identity_s456_lso10_both_nd30
stgcn_wind_aware_s123_lso10_both_nd30
stgcn_wind_aware_s42_lso10_both_nd30
stgcn_wind_aware_s456_lso10_both_nd30
Mode both — held out: site_106, site_107, site_108, site_1422, site_1427, site_1432, site_1434, site_14
Persistence
Seasonal naive (24h)
Hour-of-week average
Ridge (fit on training stations only)
Network mean (persistence)
IDW neighbour (persistence)
IDW neighbour (seasonal)
graph_wavenet_distance_s123_lso10_both
graph_wavenet_distance_s42_lso10_both
graph_wavenet_identity_s123_lso10_both
graph_wavenet_identity_s42_lso10_both
graph_wavenet_wind_aware_s123_lso10_both
graph_wavenet_wind_aware_s42_lso10_both
stgcn_distance_s123_lso10_both
stgcn_distance_s42_lso10_both
stgcn_identity_s123_lso10_both
stgcn_identity_s42_lso10_both
stgcn_wind_aware_s123_lso10_both
stgcn_wind_aware_s42_lso10_both
Mode target — held out: site_106, site_107, site_108, site_1422, site_1427, site_1432, site_1434, site_1
Persistence
Seasonal naive (24h)
Hour-of-week average
Ridge (fit on training stations only)
Network mean (persistence)
IDW neighbour (persistence)

Table 16: Leave-station-out, paired bootstrap over test weeks (A minus B; negative means A is better).

comparison	ΔMAE	95% CI	$P(A < B)$
stgcn wind aware vs stgcn identity	+0.38	[-0.05, 0.84]	0.042
stgcn distance vs stgcn identity	+0.63	[0.01, 1.29]	0.023
graph wavenet wind aware vs graph wavenet identity	+0.60	[0.09, 1.17]	0.011
graph wavenet distance vs graph wavenet identity	+1.03	[0.47, 1.60]	0.000
stgcn identity vs ridge	-12.13	[-14.11, -10.27]	1.000
stgcn identity vs seasonal naive	-6.05	[-7.53, -4.75]	1.000
stgcn distance vs idw seasonal	-5.27	[-6.51, -4.09]	1.000
stgcn identity vs idw seasonal	-5.89	[-7.14, -4.77]	1.000
stgcn identity vs network mean	-11.66	[-13.60, -9.78]	1.000
stgcn distance vs stgcn identity	-14.87	[-18.73, -11.08]	1.000
stgcn wind aware vs stgcn identity	-13.65	[-17.56, -9.85]	1.000
graph wavenet distance vs graph wavenet identity	-15.23	[-19.08, -11.55]	1.000
graph wavenet wind aware vs graph wavenet identity	-12.86	[-16.18, -9.76]	1.000
stgcn distance vs idw seasonal	+13.08	[8.68, 18.00]	0.000
stgcn distance vs idw persistence	+6.67	[2.74, 11.13]	0.000
stgcn distance vs network mean	+7.31	[3.32, 11.73]	0.000
graph wavenet distance vs idw seasonal	+14.20	[9.43, 19.55]	0.000
idw seasonal vs network mean	-5.77	[-7.39, -4.24]	1.000
stgcn distance vs stgcn identity	-23.04	[-28.08, -18.38]	1.000
stgcn wind aware vs stgcn identity	-23.69	[-28.96, -18.73]	1.000
stgcn distance vs idw seasonal	+1.08	[-0.41, 2.75]	0.082
stgcn distance vs network mean	-4.68	[-6.29, -2.99]	1.000
graph wavenet distance vs idw seasonal	+0.58	[-0.88, 2.22]	0.242

Table 17: Node dropout: the models are trained with a random 30% of stations' own history hidden in each training window, so the held-out condition is one they have practised. MAE on the held-out stations, own history hidden.

configuration	MAE	R^2
Persistence	50.73	0.269
Seasonal naive (24h)	41.40	0.479
Hour-of-week average	63.48	-0.052
Ridge (fit on training stations only)	47.48	0.378
Network mean (persistence)	47.01	0.402
IDW neighbour (persistence)	47.65	0.376
IDW neighbour (seasonal)	41.24	0.506
graph wavenet distance / lso10 both nd30	42.14	0.455
graph wavenet distance / lso10 both nd30	41.60	0.456
graph wavenet distance s456 nd30	41.72	0.467
graph wavenet identity / lso10 both nd30	65.43	-0.147
graph wavenet identity / lso10 both nd30	65.39	-0.139
graph wavenet identity s456 nd30	65.91	-0.061
graph wavenet wind aware / lso10 both nd30	42.36	0.498
graph wavenet wind aware / lso10 both nd30	41.94	0.488
graph wavenet wind aware s456 nd30	41.34	0.494
stgcn distance / lso10 both nd30	42.55	0.432
stgcn distance / lso10 both nd30	41.78	0.462
stgcn distance s456 nd30	42.64	0.430
stgcn identity / lso10 both nd30	65.38	-0.107
stgcn identity / lso10 both nd30	65.35	-0.119
stgcn identity s456 nd30	65.38	-0.107
stgcn wind aware / lso10 both nd30	42.38	0.460
stgcn wind aware / lso10 both nd30	41.33	0.490
stgcn wind aware s456 nd30	41.34	0.476

Table 18: Test performance inside and outside severe episodes (μg/m^3).

Model	Overall MAE	Episode MAE	Non-episode MAE
stgcn_identity	32.70	–	–
stgcn_wind_aware	33.09	–	–
stgcn_distance	33.24	–	–
graph_wavenet_identity	33.54	–	–
graph_wavenet_wind_aware	33.82	–	–
graph_wavenet_distance	34.69	–	–

- **GATv2-STGCN** (`src/models/gatv2_stgcn.py`) replaces the fixed adjacency matmul with multi-head GATv2 attention $\alpha_{ij} = \mathrm{softmax}_j(a^\top \mathrm{LeakyReLU}([Wh_i \,\|\, Wh_j]))$, with the adjacency acting as a sparsity mask. This tests whether learned spatial attention beats the static and wind-aware graphs.

Both were trained and evaluated on the same 8-station sample dataset that the tests use, so a reviewer can reproduce them in seconds. On that sample the GCN-GRU reaches MAE 49.65 and GATv2-STGCN 49.53 μg/m^3, both comfortably ahead of ridge (63.86) and persistence (67.09) on the same windows. These numbers are not a replication of the main result – the sample is two months of eight stations – and we report them only to show that the two additional architectures train and evaluate correctly within the same pipeline. They do not change the conclusions above, which rest on the 42-station, 7-year experiment.

6 Discussion

6.1 What the graph comparison shows

All four graph structures were trained and evaluated identically, with eight seeds each for the two graph architectures, and judged with a paired bootstrap over the 105 ISO weeks of the test period, which respects the temporal correlation in the errors in a way that a per-seed standard deviation does not. The differences that follow are statistically clear but small in absolute terms: the largest gap between graph structures is about 1.3 μg/m^3 on a score of roughly 33, i.e. around 4%, and the gap between the best and the worst configuration is about 3%. A reader should weigh the claims below accordingly: the ordering is reproducible and the confidence intervals exclude zero, but a 3% difference in PM2.5 forecasting error is not a difference that would change an operational decision on its own. The margin that does matter is against the simple baselines, and it is large: the best model beats the best simple baseline at every horizon, by 15.5% overall.

The picture is specific rather than a blanket negative. For Graph WaveNet the wind-aware graph is significantly better than the static distance graph: 33.77 against 34.60 μg/m^3, a paired difference of 0.78 with a 95% confidence interval of [0.40, 1.15]. For STGCN the same comparison is inconclusive: 33.05 against 33.18, a difference of 0.09 whose interval of [-0.11, 0.30] straddles zero. What holds for both architectures is that mixing information between stations does not pay: dropping the graph entirely beats either fixed graph by 0.38 (STGCN) and 0.33 (Graph WaveNet), whose interval [-0.03, 0.71] straddles zero. The strongest version of this claim comes from Graph WaveNet's self-adaptive adjacency, which is learned end-to-end from the data rather than designed by us: it is the worst graph of the four at 34.70, losing to no graph by 1.26 μg/m^3 (CI [0.56, 2.03]) and to the wind-aware graph by 0.94 (CI [0.50, 1.41]).

Two diagnostics turn that ranking into an explanation. The network-mean diagnostic shows that the graph is not delivering the regional signal and being ignored for it: a no-graph model handed the network-wide mean PM2.5 as an extra input channel gains nothing (0.05 μg/m^3, CI [-0.11, -0.00]) and is significantly better than every graph model. And the leave-station-out experiment shows that when a station's own history is unavailable, a distance-weighted average of the neighbours beats the graph neural network by 13.1 μg/m^3. Both point the same way: the per-station autoregressive signal is what matters, and every form of inter-station mixing we tried dilutes it.

Our answer to the research question is therefore narrow but defensible. A wind-aware graph beats a static distance graph in one of the two architectures and ties it in the other. Neither justifies mixing stations, and a learned graph does not either. Where a graph is clearly worthless is the case we went looking for it in: a station the model has never seen and whose own history is missing. There the graph is beaten by a two-line distance weighting.

The honest limitation of that answer is that we tested one construction of a wind-aware graph, against one alternative wind source. A time-lagged advection kernel, a finer wind field, or gradient-driven edge weights are different constructions, and this study does not speak to them.

A plausible reading is that the stations are too far apart, and too few, for transport to be the binding constraint. The median separation between station pairs is 16.4 km and a quarter of the pairs are more than 23 km apart; with 42 stations spread over Delhi, Noida and Ghaziabad, a 48-hour horizon is governed by the regional pollution trajectory rather than by the gradient between any two neighbouring monitors. The error budget below shows where the difficulty actually is.

6.2 A negative result that was really a bug

An earlier version of this study reported that every architecture underperformed persistence, that all R^2 values were near zero, and that graph structure was useless. That conclusion was an artefact. Two defects combined: a synthetic-data gap that made the training and validation distributions incomparable, and an output parameterisation that asked the network to predict absolute concentrations, for which the cheapest optimum is the training mean. Once the model was reparameterised to predict changes from the last observation and a linear skip was added, the same architectures moved from below-persistence to clearly above it. The lesson we take is methodological: a diagnostic that all models share a suspiciously similar R^2 is a signal that the models have not learned, and such a result should not be reported as a finding about the task.

6.3 Failure analysis

Three failure modes account for most of the residual error. First, abrupt anthropogenic spikes (Diwali, crop-residue burning) are not predictable from meteorology or neighbouring stations, and every model smooths them away. Second, sensor outages and stale readings produce flat segments that the models cannot anticipate, though the mask channel at least prevents them from being treated as real signal. Third, and most importantly, severe episodes are under-dispersed: models trained on mostly-clean conditions regress toward the conditional mean and under-predict the peaks, which is why episode error is three to five times the non-episode error for every architecture. A quantile or extremal loss, or a peak-weighted training schedule, is the obvious next step and is not attempted here.

6.4 Limitations

- Seven years of data, but the test block covers only two winters, so generalisation across pollution regimes is still only partly tested. The rolling-origin evaluation in Section 5.4 is our partial answer to this.
- Weather is measured at each station rather than taken from a reanalysis, so the wind field carries real local structure; but the per-station wind vectors agree only weakly (mean cross-station correlation of the u-component is 0.11), so the period-mean direction used to orient the graph is itself a noisy quantity.
- The wind-aware graph assumes a single prevailing direction per station; real transport is multi-directional.
- Single-direction, single-hazard forecast; the models do not condition on the previous episode's duration.
- Graph structures are static; only Graph WaveNet's adaptive variant learns edge weights.

7 Conclusion

Across seven years of real observations at 42 stations, every neural model we trained beats two strong baselines—ridge regression on the last 24 lags and persistence—once it is asked to predict the change from the last observation rather than the absolute concentration. The margin over the best simple baseline is 15.5% overall (32.66 against seasonal naive's 38.63, paired CI [4.76, 7.33]) and is positive at every horizon. The graph question has a more specific answer than a simple negative. A wind-aware directed graph does beat a static distance graph for Graph WaveNet (0.78 μg/m^3, 95% CI [0.40, 1.15] by paired bootstrap over test weeks), is indistinguishable from it for STGCN, and in both cases both are worse than using no graph structure at all—as is a graph learned end-to-end, which is the worst of the four. Two diagnostics say why. Handing a no-graph model the network-wide mean PM2.5 as an extra input feature gains nothing, so the graph is not delivering the regional signal; and on a station the model has never seen and whose own history is missing, a distance-weighted average of the neighbours beats the graph neural network by 13.1 μg/m^3. The per-station autoregressive signal is what matters, and inter-station mixing dilutes it. The remaining error is concentrated almost entirely in severe pollution episodes, where all architectures degrade sharply and where the differences between them are largest. Whether a differently constructed graph pays off on a denser network, a finer temporal resolution, or a longer and more diverse dataset is left open.

Acknowledgments

Data from CPCB (via the OpenCity and `Vonter/india-cpcb-aqi` archives). Code in the accompanying repository.

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